

TARESH GULERIA

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EDUCATION

- **Ph.D., Industrial Engineering** Dec 2026
Texas A&M University, College Station, TX, USA GPA: 4.0/4.0
- **M.S., Industrial and Systems Engineering** Dec 2024
Texas A&M University, College Station, TX, USA GPA: 3.45/4.0
- **M.Engg., Production Engineering** June 2020
Thapar Institute of Engineering and Technology, Punjab, India GPA: 8.56/10
- **B.Tech (BS), Mechanical Engineering** June 2018
SRM Institute of Science and Technology, Chennai, India GPA: 7.29/10

EXPERIENCE

- **R&D Intern – EOS of North America** Jan 2026 – May 2026
Metals R&D & Govt. Relations Pflugerville, TX, USA
 - Led R&D experimental design on superelastic NiTi components for production-ready shape memory parts.
 - DELTA-QUAL and ACCELERATE (America Makes): designing LPBF builds that cut re-qualification coupons by ~80% while meeting mechanical and microstructural targets with in-situ monitoring.
 - RECAP (America Makes): Implemented closed-loop powder reuse, improving powder utilization and reducing virgin powder consumption and waste by ~75%.
 - Headstarted VITAL Project (LIFT) on closed-loop LPBF control and property prediction to align qualification strategies across multiple OEM platforms.
- **Research Intern – Oak Ridge National Laboratory** May 2024 – Aug 2024
Manufacturing Demonstration Facility – DS&T Knoxville, TN, USA
 - Led the SMA project on LPBF pseudoelastic NiTi, focusing on anisotropic tensile and cyclic behavior driven by grain morphology and texture.
 - Performed EBSD and in-situ XRD to link process-controlled microstructure and martensitic transformation to strain localization and pseudoelastic response.
 - Processed near-fully dense NiTi by tuning LPBF parameters and mapped process–structure–property relations for deformation banding and cyclic stability.
- **Graduate Research Assistant – Texas A&M University** Jan 2023 – Present
Department of Industrial & Systems Engineering College Station, TX, USA
 - Led projects for NiTi & NiTiHf alloys using LPBF on a 3D Systems ProX 200 with optimized process windows.
 - Developed printability maps enabling defect-free processing of functionally graded NiTi-based alloys.
 - Led project with LANL on NiTi lattice and sealing ring development efforts.
 - Analyzed the effect of vector scan length and part geometry on thermal behavior using in-situ process monitoring.
 - Designed and tested Meta-Materials for NiTi-based components in collaboration with Sandia NL.
- **Graduate Engineer Trainee – Siemens** Aug 2017 – Mar 2018
Oil and Gas Sector Gurugram, HR, India
 - Implemented Building Information Modeling (BIM) for a gas-based power plant project in Berlin.
 - Designed ductile piping systems using AVEVA-PDMS and CREO for mechanical integrity and maneuverability.
 - Oil and Gas pipe wall thickness: Computational evaluations to improve stress resistance under operational loads.

PUBLICATIONS

- **T. Guleria**, A. Elsayed, K.C. Atli, R. Arroyave, I. Karaman, A. Elwany. *Beyond the material-per-function paradigm: Functionally graded active structures via laser powder bed fusion additive manufacturing of nickel-titanium shape memory alloys*. *Additive Manufacturing*, 2026. DOI: [10.1016/j.addma.2026.105276](https://doi.org/10.1016/j.addma.2026.105276)
- **T. Guleria**, A. Elsayed, A. Aga, J. Mingear, K.C. Atli, A. Elwany, I. Karaman. *Adaptive Sealing Structures with Functionally Graded NiTi Shape Memory Alloy Rings*. (Submitted to *Materials & Design*).
- **T. Guleria**, A. Elsayed, K.C. Atli, A. Elwany, I. Karaman. *Effect of Scan Length and Geometry on the Thermal History Behaviour of Melt Pools for Ni-Rich NiTi via LPBF*. (In preparation).
- **T. Guleria**, N. Verma, S. Zafar, V. Jain. *Fabrication of Kevlar[®]-reinforced UHMWPE composite through microwave-assisted compression molding for body armor applications*. *Journal of Reinforced Plastics and Composites*, 40(7-8), 307-320, 2021. DOI: [10.1177/0731684420959449](https://doi.org/10.1177/0731684420959449)
- A. Elsayed, **T. Guleria**, H. Tian, B.P. Sahu, K.C. Atli, A. Olleak, A. Elwany, R. Arroyave, D. Lagoudas, I. Karaman. *Functionally Graded NiTiHf High-Temperature Shape Memory Alloys Using Laser Powder Bed Fusion: Localized Phase Transformation Control and Multi-Stage Actuation*. *Acta Materialia*, 121175, 2025. DOI: [10.1016/j.actamat.2025.121175](https://doi.org/10.1016/j.actamat.2025.121175)
- A. Elsayed, **T. Guleria**, K.C. Atli, B. Young, P.J. Noell, B.L. Boyce, A. Elwany, R. Arroyave, I. Karaman. *Superelastic interlocking metasurfaces enabled by additive manufacturing*. *Materials Today*, 2026. DOI: [10.1016/j.mattod.2026.103369](https://doi.org/10.1016/j.mattod.2026.103369)
- A. Elsayed, **T. Guleria**, K.C. Atli, O. Bolmin, B. Young, P.J. Noell, B.L. Boyce, et al. *Active interlocking metasurfaces enabled by shape memory alloys*. *Materials & Design*, 244, 113137, 2024. DOI: [10.1016/j.matdes.2024.113137](https://doi.org/10.1016/j.matdes.2024.113137)
- P. Fernandez-Zelaia, C. Ledford, C.M. Fancher, S. Graham, **T. Guleria**, et al. *Microstructure-sensitive mechanical behavior of an additively manufactured pseudoelastic shape memory alloy*. *Additive Manufacturing Letters*, 100270, 2025. DOI: [10.1016/j.addlet.2025.100270](https://doi.org/10.1016/j.addlet.2025.100270)
- R. Lawrence, Z. He, D.K. Chakravorty, W. Brashear, H. Liu, S.B. Nite, L.M. Perez, **T. Guleria**, et al. *Cybersecurity and Data Science Curriculum for Secondary Student Computing Programs*. DOI: [10.22369/issn.2153-4136/14/2/2](https://doi.org/10.22369/issn.2153-4136/14/2/2)

PROJECTS

- **Functionally Graded Alloys and Printability Maps (NSF Grant)**: Led the development of NiTi alloys using LPBF on a 3D Systems ProX 200. Optimized parameters for NiTi 51.1 and NiTiHf, improving SME and superelasticity. Built data-driven printability maps for tailored processing.
- **Active Interlocking Metasurfaces (Sandia National Labs)**: Designed and manufactured NiTi-based interlocking arrays using LPBF. Integrated FEA and DIC for performance validation. Conducted thermomechanical testing showing high locking force, shape recovery, and cyclic stability.
- **Lattice Structures and Sealing Rings (Los Alamos National Lab)**: Leading development of NiTi lattice structures and sealing rings for radioactive material containment.
- **Composites Advanced Manufacturing (IIT Mandi)**: Developed Kevlar/UHMWPE composites using microwave-assisted compression molding for body armor.
- **Piping Standards in Oil and Gas - Siemens Energy**: Worked for Siemens to develop standards for oil and gas plant piping systems.

CONFERENCES

- **ASTM International Conference on Advanced Manufacturing 2025 - Las Vegas, NV**: Participated as a lead author on additive manufacturing of functional NiTi alloys and their mechanical behavior in advanced applications.

- **2025 Annual International Solid Freeform Fabrication Symposium - Austin, TX (SFF Symp 2025):** Co-author of the accepted abstract titled "*Mechanical Behavior of an Additively Manufactured Pseudoelastic Nitinol*". Presented research on microstructure-driven pseudoelasticity in LPBF NiTi systems.
- **International Conference on Composites Manufacturing (IIT Mandi - 2022):** Presented work on Kevlar/UHMWPE armor composites using microwave-assisted molding.

LICENSES & CERTIFICATIONS

- **Python for Data Science, AI & Development** — IBM, Issued Sept 2023.
- **Responsible Conduct of Research for Engineers** — CITI Program, Issued Jan 2023, Expires Jan 2026. Credential ID: 53414127.

JUDGEMENTS

- Served as a judge in **Student Research Week** at Texas A&M University. Evaluated 30 scientific research posters from both graduate and undergraduate students.

ACHIEVEMENTS

- **Best Poster Award – ASTM Additive Manufacturing Student Poster Session** Awarded 2nd place out of 20+ entries at the ISO/ASTM TC261-F42 Joint Meeting (March 2025).
- **NSF INTERN** - Awarded the prestigious NSF INTERN supplemental fellowship to conduct applied research at Oak Ridge National Laboratory in Summer 2024.
- **Academic Distinction – Top 5% of Class** Graduated in the top 5% of the M.Engg. Production Engineering class at Thapar Institute of Engineering and Technology (2020).

EXTRACURRICULAR ACTIVITIES

- **Technical Fest Coordinator – Mechanical Engineering (2017)** Led planning and logistics for a department fest with 3000+ attendees.
- **ISTE Event Organizer** Conducted flagship events like *Junkyard Wars* and *Infusion* to promote student engagement.
- **Interests** Hiking — Strength Training — Guitar — Running — Reading.